

```
import numpy as np
import pandas as pd
import cv2 as cv
from PIL import Image
from skimage import io
from skimage import transform
import matplotlib.pyplot as plt

urls = [
    "https://iiif.lib.ncsu.edu/iiif/0052574/full/800,/0/default.jpg",
    "https://iiif.lib.ncsu.edu/iiif/0016007/full/800,/0/default.jpg",
    "https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcReLISxRf57rninb-LEmVtDBUeXggw9tVdDMqXn_-Vep7S1YqubrHt_Nlk&s=10"
]

print("Jumlah gambar:", len(urls))
```

Jumlah gambar: 3

```
import requests
from io import BytesIO
from PIL import Image
import matplotlib.pyplot as plt

for i, url in enumerate(urls):
    try:
        response = requests.get(url, timeout=15)
        response.raise_for_status()

        image = Image.open(BytesIO(response.content)).convert("RGB")

        plt.figure(figsize=(6, 4))
        plt.imshow(image)
        plt.axis("off")
        plt.title(f"Gambar {i+1}")

        plt.show()

    except Exception as e:
        print(f"Gagal membaca gambar {i+1}")
        print("Error:", e)
```

Gambar 1



Gambar 2



Gambar 3



```
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image
import requests
from io import BytesIO

for i, url in enumerate(urls):
    try:
        response = requests.get(url, timeout=15)
        response.raise_for_status()

        image = Image.open(BytesIO(response.content)).convert("RGB")

        width, height = image.size

        new_width = width // 2
        new_height = height // 2

        resized_image = image.resize((new_width, new_height))
```

```
plt.figure(figsize=(10, 4))
```

```
plt.subplot(1, 2, 1)
plt.imshow(image)
plt.title("Gambar Asli")
plt.axis("off")
```

```
plt.subplot(1, 2, 2)
plt.imshow(resized_image)
plt.title("Gambar 50%")
plt.axis("off")
```

```
plt.show()
```

```
except Exception as e:
```

```
print(f"Gagal memproses gambar {i+1}: {e}")
```

Gambar Asli



Gambar 50%



Gambar Asli



Gambar 50%



Gambar Asli



Gambar 50%



```
image_np = np.array(image)
```

```
print("Dimensi array :", image_np.ndim)
print("Shape      :", image_np.shape)
print("Jumlah pixel  :", image_np.size)
print("Tipe data     :", image_np.dtype)
print("Ukuran elemen :", image_np.itemsize, "byte")
```

```
Dimensi array : 3
Shape      : (554, 554, 3)
```

```
Jumlah pixel : 920748  
Tipe data : uint8  
Ukuran elemen : 1 byte
```

```
# Mengambil pixel pada koordinat tertentu  
x = 100  
y = 100
```

```
pixel = image_np[y, x]  
  
print("Nilai pixel:", pixel)
```

```
Nilai pixel: [94 74 49]
```

```
image_garis = image_np.copy()  
  
height, width, channels = image_garis.shape  
  
y_tengah = height // 2  
  
image_garis[y_tengah, :, :] = [255, 255, 255]  
  
plt.figure(figsize=(8, 5))  
plt.imshow(image_garis)  
plt.title("Garis Horizontal Putih")  
plt.axis("off")  
plt.show()
```

Garis Horizontal Putih



```
image_vertikal = image_np.copy()  
  
height, width, channels = image_vertikal.shape  
  
x_tengah = width // 2  
  
image_vertikal[:, x_tengah, :] = [255, 255, 255]  
  
plt.figure(figsize=(8, 5))  
plt.imshow(image_vertikal)  
plt.title("Garis Vertikal Putih")  
plt.axis("off")  
plt.show()
```

Garis Vertikal Putih



```
image_diagonal = image_np.copy()

height, width, channels = image_diagonal.shape

for i in range(min(height, width)):
    image_diagonal[i, i] = [255, 255, 255]

plt.figure(figsize=(8, 5))
plt.imshow(image_diagonal)
plt.title("Garis Diagonal Putih")
plt.axis("off")
plt.show()
```

Garis Diagonal Putih



```
image_horizontal = image_np.copy()

height, width, channels = image_horizontal.shape

y = height // 2

panjang_garis = 300

x_awal = (width - panjang_garis) // 2
x_akhir = x_awal + panjang_garis

image_horizontal[y, x_awal:x_akhir] = [255, 255, 255]

plt.figure(figsize=(8, 5))
plt.imshow(image_horizontal)
```

```
plt.title("Garis Horizontal Putih - Panjang 300 Pixel")  
plt.axis("off")
```

Garis Horizontal Putih - Panjang 300 Pixel



```
image_kotak = image_np.copy()  
  
height, width, channels = image_kotak.shape  
  
# Koordinat kotak  
x1 = width // 4  
y1 = height // 4  
  
x2 = x1 + 200  
y2 = y1 + 150  
  
# Isi kotak dengan warna putih  
image_kotak[y1:y2, x1:x2] = [255, 255, 255]  
  
plt.figure(figsize=(8, 5))  
plt.imshow(image_kotak)  
plt.title("Kotak Putih Terisi Penuh")  
plt.axis("off")  
plt.show()
```

Kotak Putih Terisi Penuh

